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Block 1: Parametric and Non-Parametric Statistics

Unit 1: Parametric and Non-Parametric Statistics

- **1.0 Introduction**
    - **Parametric statistical tests** specify certain conditions about the data, such as it being normally distributed.
    - **Non-parametric statistics** do not require such stringent conditions and are known as **distribution-free tests**.
- **1.2 Definition of Parametric and Non-parametric Statistics**
    - **Parametric tests** typically involve data expressed in absolute numbers or values, like Student's t-test. They operate under certain assumed conditions, and the meaningfulness of their results depends on these assumptions being valid.
    - **Non-parametric tests** are statistical tests not based on a normal distribution of data or any other specific assumption. The data are generally **ranked or grouped**.
        - Examples include the **chi-square test** and **Spearman's rank correlation coefficient**.
        - They do not rely on data belonging to any particular distribution, making them **distribution-free methods**.
        - They are often based on the ranks of observations and do not depend on any population distribution.
        - They do not assume a fixed model structure, and assumptions of classical tests are not applied.
- **1.3 Assumptions of Parametric and Non-parametric Statistics**
    - **1.3.1 Assumptions of Parametric Statistics**
        - The population from which the sample is drawn should be **normally distributed**.
        - Variables must be measured on an **interval or ratio scale**.
        - Observations must be **independent**; the selection of one case shouldn't affect others.
        - Populations must have the **same variance (homoscedasticity)** or a known ratio of variance. This is particularly important for small samples.
    - **1.3.2 Assumptions of Non-parametric Statistics**
        - Used when assumptions for parametric tests **cannot be met**.
        - Advisable when sample size is **very small (N ≤ 30, e.g., N=5 or N=6)**.
        - When assumptions like **normality of distribution are doubtful**.

        - When data is in **ordinal or nominal scale**, or expressed in ranks, or as +/– signs.
        - When the nature of the **population is not known to be normal**.
        - Fewer and weaker assumptions compared to parametric tests (e.g., observations are independent, underlying continuity).
- **1.4 Advantages of Non-parametric Statistics**
    - Often the **only alternative for very small sample sizes**.
    - Make **fewer assumptions** about the data.
    - Hypotheses tested may be **more appropriate for research**.
    - Can analyze data inherently in **ranks or categorical/classificatory (nominal) scale**.
        - *Example:* Studying anxiety where one can only state subject A is more anxious than B, without exact measurement.
    - Can handle samples from **several different populations** which parametric tests sometimes cannot.
    - Generally **easier to learn and apply**, with more direct interpretation.
    - **Assumption-free** and economical in time.
- **1.5 Disadvantages of Non-parametric Statistical Tests**
    - **Wasteful** if all parametric assumptions are met (less power-efficient). A 90% power-efficiency means the parametric test is as effective with a 10% smaller sample.
    - Tables for implementation can be **scattered and in different formats**.
    - Less precise and have **less power** than parametric tests.
- **1.6 Parametric Statistical Tests for Different Samples**
    - **Population**: The entire group a researcher intends to understand.
    - **Sample**: A small proportion of the population chosen to represent it.
    - **Central Limit Theorem**: If large (N > 30) random samples are drawn from an infinite population:
        - Sample means will be **normally distributed**.
        - Average of sample means will be the **population mean**.
        - Distribution of sample means has its own **standard deviation**, called the **standard error of the mean (SEM or σM)**.
            - SEM indicates how far sample means might deviate from the population mean.
            - Formula for large sample: $\sigma M = \sigma / \sqrt{N}$.
    - For **small samples (N ≤ 30)**, the sampling distribution of means is **not normal**; it follows a **t-distribution**.
- **1.7 Parametric Statistical Measures for Calculating the Difference Between Means**
    - Used to determine if a **true difference exists between population parameters of two samples** (means, SDs, correlations).

    - **1.7.1 Significance of Difference Between Means of Two Independent Samples**
        - Means are independent if from samples drawn from **totally different/unrelated groups**.
        - For large samples, the distribution of differences between means tends to be normal with a mean of zero.
        - **Standard Error of the Difference of Means (σdM)**: **σdM = √(σM1² + σM2²)**.
    - **1.7.2 Significance of Difference Between Means of Two Dependent Samples**
        - Means are dependent if from **same sample on two occasions** or matched equivalent samples.
    - **1.7.3 Significance of Difference Between Means of Three or More Samples**
        - Requires computing **F-value** via **Analysis of Variance (ANOVA)**.
        - **Assumptions for ANOVA**: Population distribution should be **normal** (though robust to deviations), groups must be **randomly chosen**, and sub-groups should have **same variability (homogeneity of variances)**, tested by **Fmax test**.
- **1.8 Parametric Statistics Measures Related to Pearson's 'r'**
    - Sampling distribution of Pearson's 'r' is **not normal** unless population 'r' is near zero and sample size is large (N≥30).
    - For inference regarding 'r' (especially high/low values), convert 'r' to **Fisher's Z coefficient** (which has a normal sampling distribution regardless of N or population 'r').
    - Formula for Standard Error of Z (σz): **SEz = 1/√(N–3)**.
- **1.8.1 Non-parametric Tests Used for Inference (Unrelated/Independent Samples)**
    - **Chi-square (X^2)** test:
        - Applied to **discrete (counted) data**.
        - Test of **independence**; estimates likelihood observed relationship is due to factors other than chance.
        - **Not a measure of degree of relationship**.
        - Assumes random sample observations.
        - Formula: **$X^2 = \Sigma [(fo-fe)^2 / fe]$**.
    - **Median test**: Tests if two independent samples differ in central tendencies (same median). Useful for ordinal data.
    - **Mann-Whitney U test**: More useful alternative to parametric t-test when assumptions are not met, and data is ordinal.
- **1.9 Some Non-parametric Tests for Related Samples**
    - **Sign Test**:
        - Simplest significance test.
        - Uses **plus/minus signs** instead of quantitative measures.

        - Useful when quantitative measurement is difficult but performance can be ranked (superior/inferior).
        - Assumes variable has a **continuous distribution**.
    - **Wilcoxon Matched-Pairs Signed-Ranks Test**:
        - **More powerful than sign test**.
        - Tests **direction and magnitude of differences** within matched pairs.
        - Deals with **dependent groups** (matched pairs), not independent groups.

Unit 2: Descriptive and Inferential Statistics

- **2.0 Introduction**

    1. This unit deals with **descriptive and inferential statistics**.
- **2.2 Meaning of Descriptive Statistics**

    1. **Statistics** in singular sense refers to **statistical methods** (collection, classification, analysis, interpretation of data). In plural, it refers to 'data'.
    2. **Descriptive Statistics**: A branch of statistics dealing with **describing obtained data**. It defines a particular group/population for corresponding characteristics.
        - Includes: **classification, tabulation, diagrammatic/graphical presentation, measures of central tendency, and variability**.
        - Goal: **summarize observations into a single estimate (parameter)**.
    3. Involves two operations: **Organization of data** and **Summarization of data**.
- **2.3 Organization of Data**

    1. Major techniques: **Classification, Tabulation, Graphical Presentation, Diagrammatical Presentation**.
    2. **2.3.1 Classification**: Arranging data into groups based on similarities.
        - **Frequency Distribution**: Shows number of cases within a class interval/range of scores.
            - **Ungrouped Frequency Distribution**: Lists all score values and their tallies/frequencies.
            - **Grouped Frequency Distribution**: Organizes data into classes, useful for wide score ranges.
            - **Class Intervals**: Determined by range, number of classes (5-30), and uniform width (i).
            - **Class Limits Methods**:
                - **Exclusive method**: Upper limit of one class is lower limit of next (e.g., 30-40, 40-50); upper limit is excluded from the class.
                - **Inclusive method**: Upper limit of one class is not lower limit of next (e.g., 30-39, 40-49); upper limit is included.
                - **True or Actual class method**: Scores extend 0.5 units below to 0.5 units above face value (e.g., 29.5 to 39.5).

        - **Types of Frequency Distribution**: Relative, Cumulative, Cumulative relative.
    3. **2.3.2 Tabulation**: Presenting classified data in a table (rows and columns).
        - Components: Table number, Title, Caption (column headings), Stub (row headings), Body, Head note, Footnote, Source.
    4. **2.3.3 Graphical Presentation**: Displaying frequency distribution using graphs (X and Y axes).
        - **Histogram**: Series of rectangles, width = class interval, height = frequency (for continuous data).
        - **Frequency Polygon**: Line graph connecting midpoints of class intervals at their frequencies.
        - **Frequency Curve**: Smooth freehand curve through a frequency polygon.
        - **Cumulative Frequency Curve or Ogive**: Graph of cumulative frequency distribution.
            - **'Less than' Ogive**: Cumulative frequencies plotted against upper class boundaries (increasing curve).
            - **'More than' Ogive**: Cumulative frequencies plotted against lower class boundaries (decreasing curve).
    5. **2.3.4 Diagrammatic Presentation**: Visual form for statistical data, simpler than graphs.
        - Bar diagram, Sub-divided bar diagram, Multiple bar diagram, Pie diagram, Pictogram.
- **2.4 Summarisation of Data**

    1. **2.4.1 Measures of Central Tendency**: Middle point of a distribution; single estimate summarizing observations.
        - Characteristics of a good measure: Clearly defined, comprehensible, easy to compute, based on all observations, amenable to math treatment, least affected by sampling fluctuation.
        - **Arithmetic Mean**: Most popular average (sum of values / number of values).
        - **Median**: Middle-most value in an ordered distribution; divides into two equal parts; not affected by extreme values.
        - **Mode**: Value with maximum frequency; most typical value.
    2. **2.4.2 Measures of Dispersion**: Indicate how far observations scatter from each other or from the mean.
        - **Range**: Difference between largest and smallest scores.
        - **Average Deviation**: Arithmetic mean of differences between each score and the mean.
        - **Standard Deviation (SD)**: Most stable index of variability; square root of variance; based on all observations, amenable to math treatment, least affected by sampling fluctuation.
    3. **2.4.3 Skewness and Kurtosis**

        - **Skewness**: Degree of asymmetry in a distribution (concentration at one end).
        - **Kurtosis**: 'Peakedness' or flatness of a frequency distribution curve compared to normal.
            - **Leptokurtic**: More peaked than normal (thin distribution).
            - **Platykurtic**: More flat than normal.
            - **Mesokurtic**: Normal curve.
- **2.5 Meaning of Inferential Statistics**

    1. **Inferential Statistics**: Process of **drawing conclusions about a large group (population) based on observations from a sample**.
        - *Example:* Exit polls predicting election results from a sample of voters.
    2. Provides tools to compute probabilities of future behavior.
    3. Two types of inferential procedures: **Estimation** and **Hypothesis testing**.
- **2.5.1 Estimation**

    1. Inference about population characteristics based on sample discoveries.
    2. **Characteristics of Good Estimator**: Unbiased (mean of statistic equals parameter), Consistent (probability of estimate being close to parameter increases with sample size), Accuracy (closeness to true value depends on representativeness, sample size, variability).
    3. **2.5.2 Point Estimation**: Estimating a parameter as a **single value** (e.g., population mean $\mu = 45.0$ from sample mean $x = 45.0$).
    4. **2.5.3 Interval Estimation**: Constructing a **range (confidence interval)** of scores expected to include the population parameter (e.g., 95% or 99% confidence level). Lower and upper scores are called **confidence limits**.
- **2.6 Hypothesis Testing**

    1. Closely tied to inferential statistics.
    2. Determines if observed sample results support a particular theory or innovation for a population.
    3. **Statistical Hypothesis**: A statement (true/false) about a population parameter or its probability distribution, validated using sample information.
    4. **Null Hypothesis (H0)**: Statement of **no difference**; assumed true until proven otherwise. *Example: Two samples came from the same population; new drug is no better than current drug*.
    5. **Alternative Hypothesis (H1)**: Counter-proposition to null hypothesis; states a **significant difference** exists. *Example: Two samples belong to different populations; new drug is better*.
    6. H1 is not directly tested; acceptance/rejection is determined by H0.
    7. If **probability 'p' of H0 being correct is too low, H0 is rejected and H1 is accepted** (result is statistically significant).
    8. If 'p' is high, H0 is accepted (difference is due to chance, not significant).

- **2.6.2 Level of Significance (alpha, α)**:

    1. The probability of **chance occurrence of observed results up to and below which the null hypothesis is considered too low** and results are significant.
    2. Common choices: **0.05 (5%)** or **0.01 (1%)**.
        - **0.05 level**: Results are significant if 'p' < .05; meaning, 5 or less times out of 100 trials, observed results may arise by random sampling chance. Implies **95% confidence**.
        - **0.01 level**: Results are significant if 'p' < .01; meaning, 1 or less time out of 100 trials, observed results may arise by random sampling chance. Implies **99% confidence**.
    3. Lower alpha means smaller chance of Type I error.
- **2.6.3 One-Tail and Two-Tail Test**

    1. **One-tailed (directional) test**: Alternative hypothesis specifies a **direction** of difference (e.g., A > B or A < B). Rejection region is in **one tail** of the comparison distribution.
    2. **Two-tailed (non-directional) test**: Alternative hypothesis predicts an effect but **not a specific direction** (e.g., A ≠ B). Rejection region is in **both tails** of the comparison distribution.
- **2.7 Errors in Hypothesis Testing**

    1. Decisions based on sample data, so errors are possible.
    2. **Decision errors** are situations where correct procedures lead to wrong conclusions.
    3. **2.7.1 Type I Error (α)**:
        - **Rejecting the null hypothesis when it is actually true**.
        - Concluding study supports research hypothesis when it's false.
        - **Probability of Type I error is α (alpha)**.
        - Considered more serious in many cases (e.g., wrongly concluding new drug is better).
        - **"Producer's risk"**: Rejecting a good product lot.
    4. **2.7.2 Type II Error (β)**:
        - **Accepting (or failing to reject) the null hypothesis when it is actually false**.
        - Concluding results are inconclusive when research hypothesis is true.
        - **Probability of Type II error is β (beta)**.
        - **"Consumer's risk"**: Accepting a bad product lot.
    5. **2.7.3 Relationship between Type I and Type II Errors**:
        - **Inversely related for a fixed sample size (n)**: Reducing α increases β, and vice-versa.
        - **Only way to reduce both α and β simultaneously is to increase sample size**.

- **2.8 General Procedure for Testing a Hypothesis**

    1. **State the null hypothesis (H0)** suitable to the problem.
    2. **Define the alternative hypothesis (H1)**.
    3. **Set the criteria for a decision (Level of Significance α and Critical Region)**.
    4. **Calculate the suitable test statistics** (e.g., z-score, t-value, F-value).
    5. **Define degrees of freedom** for the test situation.
    6. **Find the probability level 'p'** corresponding to the calculated test statistic from relevant tables.
    7. **Reject or accept (fail to reject) null hypothesis** based on calculated vs. tabulated values at the chosen significance level.
        - If calculated value is in the critical region ($p < \alpha$), reject H0.
        - **Rejecting H0 does not mean it is disproved, only that sample doesn't support it. Accepting H0 doesn't mean it is proved, only that it is supported by the sample**.

Unit 3: Type I and Type II Errors

(This unit is a more detailed discussion of concepts introduced in Unit 2, particularly hypothesis testing and errors.)

- **3.2.1 Hypothesis Testing**: A systematic procedure for deciding if research study results support a theory applicable to a population.

- **3.2.2 The Core Logic of Hypothesis Testing**:

    - Researchers first spell out what would happen if the **null hypothesis (no effect/no difference) were true**.
    - Then, they find the **probability of getting the observed research results if the null hypothesis were true**.
    - If this probability is **very low**, they **reject the null hypothesis** and accept the research (alternative) hypothesis. If not low, the study is inconclusive.
    - **Comparison Distribution**: The distribution representing the population situation if the null hypothesis were true.
    - **Cutoff Sample Score (Critical Value)**: The extreme sample score needed to decide to reject the null hypothesis.
- **3.2.4 Implications of Rejecting or Failing to Reject the Null Hypothesis**:

    - **Rejecting H0**: Results are **statistically significant**; supports the research hypothesis. Do not say it "proves" it.
    - **Not rejecting H0**: Results are **not statistically significant**; the study is **inconclusive**. Do not say it "supports" the null hypothesis. It is always possible there's a small difference the study couldn't detect.

- **3.2.5 One-Tailed and Two-Tailed Hypothesis Tests**: (Detailed explanation as in Unit 2, but re-emphasized here)
    - **Directional hypotheses** lead to **one-tailed tests** (e.g., vitamin causes babies to walk *earlier*).
    - **Non-directional hypotheses** lead to **two-tailed tests** (e.g., social skills program *changes* productivity, without specifying improve or hurt).
- **3.2.6 Decision Errors**: (Detailed explanation as in Unit 2, with emphasis on the probability aspect)
    - Errors occur because decisions about populations are based on samples, and probabilities are involved.
- **3.3 Type I Error**:
    - **Rejecting H0 when H0 is true**. Concluding research hypothesis is supported when it's false.
    - **Probability of Type I error = Significance Level (α)**.
    - Serious concern for psychologists due to potential for mistaken theories/applications.
    - Lower α means smaller chance of Type I error.
- **3.4 Type II Error**:
    - **Failing to reject H0 when H0 is false**. Concluding results are inconclusive when research hypothesis is true.
    - **Probability of Type II error = β (beta)**.
    - Occurs when a true research hypothesis doesn't yield an extreme enough result to reject H0.
- **3.5 Relationship between Type I and Type II Errors**:
    - **Protecting against one type of error increases the chance of making the other**.
    - Setting a very stringent α (e.g., .001) protects against Type I error but increases Type II error risk.
    - Setting a lenient α (e.g., .20) protects against Type II error but increases Type I error risk.
    - **Increasing sample size (n) is the only way to reduce both errors simultaneously**.

Unit 4: Setting Up the Levels of Significance

- **4.2 Hypothesis Testing**:
    - Procedure to distinguish if differences in sample data are due to **chance elements** or are **significant (real)**.

    - Involves comparing competing claims: **Null hypothesis (H0)** vs. **Alternative hypothesis (H1)**.
    - Priority is given to H0; it is rejected only if evidence against it is sufficiently strong.
    - Hypotheses are often statements about **population parameters** (e.g., mean height) or **distributional forms** (e.g., normally distributed).
- **4.3 Null Hypothesis (H0)**:
    - Represents a theory believed true or used as argument basis, not yet proven.
    - Example: "There is no difference between two drugs in treating a disorder".
    - **Alternative Hypothesis (H1)**: Statement set up to be established; accepted if H0 is rejected.
    - Conclusion is always in terms of H0: "reject H0 in favour of H1" or "do not reject H0".
    - "Do not reject H0" does not mean H0 is true, just insufficient evidence against it.
- **4.4 Errors in Hypothesis Testing**: (Reiterates Type I and Type II errors with further context)
    - **Type I error**: Rejecting H0 when true (P(Type-I error) = α). **"Producer's risk"**.
    - **Type II error**: Accepting H0 when false (P(Type-II error) = β). **"Consumer's risk"**.
    - **p-value**: Measure of how likely sample results are, assuming H0 is true. If p-value < α, reject H0. Also called **observed level of significance**.
- **4.5 Confidence Limits**:
    - The **range/limits within which the hypothesis should lie with specified probabilities**.
    - Customarily set at **5% (.05) or 1% (.01) levels of significance**.
    - If sample value falls within limits, H0 is accepted; if not, H0 is rejected.
    - **Levels of Confidence**: Directly relate to significance levels.
        - **0.05 level (95% confidence)**: If experiment repeated 100 times, only 5 occasions mean will fall outside **μ ± 1.96 SE**.
        - **0.01 level (99% confidence)**: If experiment repeated 100 times, only 1 occasion mean will fall outside **μ ± 2.58 SE**.
        - **0.01 level is more rigorous**; if significant at .01, it's automatically significant at .05.
    - **4.5.2 Application and Interpretation of Standard Error of the Mean (SEM) in Small Samples**:
        - Uses **(N-1) in SD formula**.
        - Uses **t-distribution** instead of normal distribution.
    - **4.5.3 The Standard Error of a Median (σMdn)**: Variability of sample medians is ~25% greater than means in normal population. **σMdn = 1.253 * (σ / √N)**.
- **4.6 Setting Up the Level of Confidence or Significance**:

    - Experimenter decides α (e.g., 0.05 or 0.01) **before collecting data** and must adhere to it.
    - **4.6.1 Size of the Sample**:
        - **Large samples (N ≥ 30)**: Distribution of differences between means is **normal**; use z-scores (1.96 for .05, 2.58 for .01).
        - **Small samples (N < 30)**: Distribution is **t-distribution**; use t-values with **df = N1 + N2 - 2**.
    - **4.6.2 Two-tailed and One-tailed Tests of Significance**: (Reiterates Unit 2 & 3 content on directional vs. non-directional tests).
        - **Two-tailed**: Interested in **any significant difference**, regardless of direction. Employs both sides of distribution.
        - **One-tailed**: Primary concern is with the **direction of difference** (e.g., gain in skill).
- **4.7 Steps in Setting up the Level of Significance**: (Summary of Hypothesis Testing Procedure, with emphasis on alpha and critical region).
    - **Critical Region**: Extreme sample values unlikely if H0 is true; boundaries set by alpha.
    - **Test Statistic (z)**: **z = (x - μ) / σx**.
    - **4.7.1 Formulating Hypothesis and Stating Conclusions**:
        - State H1 first (research prediction).
        - H0 is the opposite of H1 and includes an equality sign.
        - If sample supports H1, reject H0 (risk α error).
        - If not enough evidence for H1, conclude H0 cannot be rejected (may need more data).
        - Analogy: Courtroom trial (defendant innocent H0, guilty H1; Type I error is convicting innocent).

Block 2: Correlation and Regression

Unit 1: Product Moment Coefficient of Correlation

- **1.0 Introduction**
    - **Correlation**: A description of "relationship" or "association" between two variables.
- **1.2 Correlation: Meaning and Interpretation**
    - **Correlation Coefficient**: A number, usually 'r' (Pearson's), expressing the association between two variables. Range is **-1.00 to +1.00**.
        - Value of 1 indicates **perfect relationship** (+1 or -1).
        - Closer to ±1, **stronger the relationship**.
        - Closer to 0, **weaker the relationship**.

- **1.2.1 Scatter Diagram (Scatterplot)**: Graphical presentation of relationship by plotting pairs of values on a graph.
  - *Example:* Plotting intelligence vs. reasoning task scores.
- **1.2.2 Correlation: Linear and Non-Linear Relationship**
  - **Linear Relationship**: Can be plotted as a **straight line** ($Y = \alpha + \beta X$). *Example:* Intelligence and Scores on Reasoning Task.
  - **Non-linear (Curvilinear) Relationships**: Cannot be plotted as a straight line. *Example:* Stress and performance (Yorkes-Dodson Law), Steven's Power Law. **Pearson's 'r' is for linear relationships**.
- **1.2.3 Direction of Correlation: Positive and Negative**
  - **Positive Correlation**: Variables move in the **same direction** (as one increases, other increases; as one decreases, other decreases). *Example:* Intelligence and marks obtained.
  - **Negative Correlation**: Variables move in **opposite directions** (as one increases, other decreases). *Example:* Intelligence and errors on reasoning task.
  - **Zero Correlation (No Relationship)**: No association between variables (correlation coefficient is zero). *Example:* Shoe size and intelligence.
- **1.2.4 Correlation: The Strength of Relationship**
  - Strength is indicated by how close 'r' is to ±1.00.
  - **Shared Variance (Coefficient of Determination, r^2)**: Percentage of common variance between X and Y is **$r^2 \times 100$**.
    - *Example:* If $r = 0.50$, shared variance = $0.50^2 \times 100 =$ **25%**. If $r = -0.81$, shared variance = $(-0.81)^2 \times 100 =$ **65.61%**.
    - **Correlation coefficient is NOT a percentage**, but its square can be interpreted as shared variance percentage.
- **1.2.5 Measurements of Correlation**
  - Sample correlation coefficient is 'r'; population correlation coefficient is **ρ (rho)**.
  - **Pearson's Product Moment Correlation (r)**: Most popular; for data on **at least interval or ratio scale**.
  - Other correlations: Spearman's Rank Order Correlation (ρs or rs) for rank-order data, Kendall's tau (τ), Biserial, Point Biserial, Tetrachoric, Phi coefficient.
- **1.2.6 Correlation and Causality**: **Correlation does not necessarily imply causality**. If X and Y are correlated, X could cause Y, Y could cause X, or a third variable Z could cause both.

- **1.3 Pearson's Product Moment Coefficient of Correlation**
  - Developed by Karl Pearson.
  - **1.3.1 Variance and Covariance: Building Blocks of Correlations**
    - **Mean (X)**: Sum of scores divided by number of observations.
    - **Variance (S^2X)**: Sum of squares of deviations of each score from mean, divided by n.

        - **Covariance (CovXY or SXY)**: Indicates association between two variables; sum of (X-X)(Y-Y) divided by n.
    - **1.3.2 Equations for Pearson's Product Moment Coefficient of Correlation**
        - **r = CovXY / (SX * SY)**.
        - Alternatively: **r = Σ(X-X)(Y-Y) / (n * SX * SY)**.
        - Sign of 'r' depends on sign of CovXY.
    - **1.3.3 Numerical Example**: (Calculation demonstrated in source for BHS and BDI scores, resulting in r = +0.937).
    - **1.3.4 Significance Testing of Pearson's Correlation Coefficient**
        - Required when 'r' is used as an **estimate of population correlation (ρxy)**.
        - **Null Hypothesis (H0): ρxy = 0** (No correlation in population).
        - **Alternative Hypothesis (HA): ρxy ≠ 0** (Correlation exists in population).
        - Uses **t-distribution** with **df = n - 2**.
        - **Two-tailed test** for non-directional HA; **one-tailed test** for directional HA.
        - **Comparison**: If obtained 'r' (absolute value) is > critical value from table, reject H0.
    - **1.3.5 Adjusted r (radj)**:
        - Sample 'r' is a **biased estimate of population 'ρ'** for small sample sizes.
        - **Adjusted r** reduces this bias: **radj = $\sqrt{(1 - [(1 - r^2) * (n - 1) / (n - 2)])}$**.
    - **1.3.6 Assumptions for Significance Testing**
        - **Independence** among pairs of scores.
        - Population of X and Y follow **normal distribution**, and pair of scores has **normal bivariate distribution**.
        - Pearson's 'r' is a **robust statistic**, meaning small violations of assumptions don't significantly impact distributional properties.
    - **1.3.7 Ramifications in the Interpretation of Pearson's r**
        - **Direction**: Positive or negative.
        - **Strength**: Calculated as percentage of shared variance (r^2).
        - **Factors that obscure interpretation**: **Restricted range, outliers, unreliability of measurement, curvilinearity**.
    - **1.3.8 Restricted Range**: If data for a variable covers only a small part of its full range, the correlation coefficient is likely to be **reduced**, leading to an invalid interpretation.
    - **1.4 Unreliability of Measurement**: Low reliability of variables **reduces correlation**; can be adjusted by calculating disattenuated correlations.
    - **1.4.1 Outliers**: Extreme scores can **drastically affect** the correlation value, altering its strength and direction.
    - **1.4.2 Curvilinearity**: Pearson's 'r' is appropriate only for **linear relationships**. If relationships are curvilinear, other techniques like Spearman's rho (if monotonic) or curve-fitting are needed.
- **1.5 Using Raw Score Method for Calculating r**
    - An alternative computational method to the deviation scores formula.

- Formula: **$r = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{[(n\Sigma X^2 - (\Sigma X)^2) * (n\Sigma Y^2 - (\Sigma Y)^2)]}$**.

Unit 2: Other Types of Correlation (Phi-Coefficient)

- **2.2 Special Types of Correlation**
    - Used when Pearson's 'r' assumptions (continuous variables, normality, linearity) are not met.
    - **Measures of Correlation**: Variables can be assigned order (e.g., number of friends).
    - **Measures of Association**: For categorical or nominal variables (e.g., gender, property ownership).
    - Types covered: Point-Biserial, Phi coefficient (Pearson's modified); Biserial, Tetrachoric (non-Pearson); Spearman's rho, Kendall's tau (rank-order).
- **2.3 Point Biserial Correlation (rPB)**:
    - Pearson's 'r' between **one truly dichotomous variable** (e.g., male-female, yes-no, experimental-control) and **one continuous variable**.
    - **Algebraically, rpb = r** (can be calculated using Pearson's formula).
    - **Interpretation of sign**: Depends on coding (e.g., 1 for female, 0 for male; positive rpb means group coded 1 has higher mean on continuous variable).
    - **Strength**: rpb^2 gives percentage of shared variance.
    - **Significance Testing of rPB**: Similar to Pearson's 'r'; uses **t-distribution** with **df = n - 2**.
- **2.4 Phi Coefficient (φ)**:
    - Pearson's 'r' when **both variables are dichotomous**. *Example:* Gender and property ownership (both 0 or 1).
    - **Interpretation of sign**: Depends on coding, but once established, indicates relationship (e.g., negative φ could mean males (0) have more ownership than females (1)).
    - **Strength**: φ^2 gives percentage of shared variance.
    - **Significance Testing of Phi (φ)**: Uses **Chi-Square (χ^2) distribution**.
        - **$\chi^2 = n\varphi^2$** with **df = 1**.
- **2.5 Biserial Correlation (rb)**:
    - Used when one variable is **measured dichotomously but has an underlying continuous distribution** (e.g., "clinically low mood" vs. "normal mood," where mood is continuous).
    - Formula relates to rpb: **$rb = rpb * \sqrt{(P0 * P1)} / h$** (where h is ordinate of normal distribution).
- **2.6 Tetrachoric Correlation (rTET)**:

    - Used when **both dichotomous variables have underlying continuous distributions** (e.g., attitude towards females and attitude towards liberalization, measured as positive/negative).
    - Formula involves cosine function based on a 2x2 contingency table.
- **2.7 Rank Order Correlations**:
    - Applicable when data is in **rank-order format** (ordinal scale) or when Pearson's assumptions are not met.
    - Also used for **monotonic relationships** (where X consistently increases Y, or consistently decreases Y, but not necessarily linearly). Not for non-monotonic relationships like Yorkes-Dodson Law.
- **2.8 Spearman's Rank Order Correlation or Spearman's Rho (rs)**:
    - Developed by Charles Spearman; suitable for **rank-order data**. Can be used with continuous data converted to ranks, or when Pearson's assumptions are violated. Assesses **monotonic relationships**.
    - **Range**: **-1.00 to +1.00**.
    - **Interpretation**: Based on sign and value (closer to ±1.00 means stronger association). Does not imply causation.
    - **Relationship to Pearson's r**: Pearson's 'r' on ranks of X and Y equals Spearman's rho.
    - **Null Hypothesis (H0): $\rho s = 0$** (no Spearman correlation in population).
    - **Alternative Hypothesis (HA): $\rho s \neq 0$** (or directional, <0, >0).
    - **Computational Formula (Untied Ranks)**: **$rs = 1 - [6\Sigma D^2 / (n(n^2 - 1))]$**. D is the difference between ranks.
    - **Tied Ranks**: When two or more subjects have the same score, they get the average of the ranks they would have occupied. Simpler procedure for tied ranks uses Pearson's formula on the ranks.
    - **Significance Testing of Spearman's rho**:
        - Refer to **Appendix D** for critical values based on 'n' and alpha level.
        - If 'n' > 10, **t-distribution** can be used with **df = n - 2**: **$t = rs * \sqrt{((n - 2) / (1 - rs^2))}$**.
- **2.9 Kendall's Tau (τ)**:
    - Another measure for **rank-order data**, alternative to Spearman's rho.
    - **Range**: **-1.00 to +1.00**.
    - **Interpretation**: Based on sign and value (closer to ±1.00 means stronger).
    - **Logic**: Based on **concordance (nC)** (pairs of ranks in agreement) and **discordance (nD)** (pairs not in agreement).
    - **Formula**: **$\tau = (nC - nD) / [n(n - 1) / 2]$**.
    - **Interpretation Example**: If $\tau = 0.778$, probability that randomly sampled subjects have similar order on two variables is 0.778 higher than probability of reverse order.
    - **Significance Testing of τ**:

        - Refer to **Appendix E** for critical values.
        - Can use **z-transformation**: **$z = \tau / \sqrt{[2(2n + 5)] / [9n(n - 1)]}$**.
        - Kendall's tau is considered better than Spearman's rho with tied ranks, and sometimes better than Pearson's 'r' with extreme non-normality.

Unit 3: Partial and Multiple Correlations

- **3.0 Introduction**
    - Correlations can involve **more than two variables** by statistically controlling for other influences.
    - **Partial correlation**: Controls for influence of a variable on *both* variables being correlated.
    - **Part (Semipartial) correlation**: Controls for influence of a variable on *only one* of the variables being correlated.
    - **Multiple Correlation**: Finds correlation between one variable and a linear combination of multiple other variables.
- **3.2 Partial Correlation (rP)**:
    - Correlation between two variables (A and B) where the **influence of one or more other variables (C)** has been **partialled out or controlled for from both A and B**. Represented as rAB.C.
    - Can be considered correlation between two sets of **residuals** (errors after removing common influence).
    - **Formula (for controlling one variable C)**: **$rAB.C = (rAB - (rAC * rBC)) / \sqrt{((1 - rAC^2) * (1 - rBC^2))}$**.
    - **Numerical Example**: Calculating partial correlation between academic achievement and anxiety, controlling for intelligence.
    - **Significance Testing of Partial Correlation**:
        - **Null Hypothesis (H0): $\rho P = 0$** (population partial correlation is zero).
        - Uses **t-distribution** with **df = n - v** (n=sample size, v=total number of variables). **$t = rP * \sqrt{((n - v) / (1 - rP^2))}$**.
    - **Alternative Use of Partial Correlation**: Can assess if groups influence correlation between two variables (e.g., gender's influence on neuroticism and intolerance to ambiguity correlation).
- **3.3 Linear Regression (Brief overview as prerequisite)**
    - Goes beyond correlation to **predict values** of one variable (Y) from another (X) by creating an **equation (regression equation)**.
    - **Regression line**: The "best fitting line" that minimizes the distance between predicted and actual values in a scatter plot.
    - **$Y = \alpha + \beta X + \varepsilon$** (Population equation) or **$Y = a + bX + e$** (Sample equation).
        - Y = dependent/criterion variable. X = independent/predictor variable.
        - α/a = Y-intercept (regression constant). β/b = slope (regression coefficient).

        - ε/e = error/residual (unexplained variance).
- **3.4 Part Correlation (Semipartial correlation, rsp)**:
    - Correlation between two variables where the influence of a third variable (C) is **partialled out from only one variable** (e.g., B) and NOT from both. Represented as rA(B.C).
    - **Formula**: **$r_{A(B.C)} = (r_{AB} - (r_{AC} * r_{BC})) / \sqrt{(1 - r_{BC}^2)}$**.
    - **Numerical Example**: Calculating semipartial correlation of anxiety (A) with academic achievement (B), partialling intelligence (C) only from academic achievement.
    - **Significance Testing of rsp**:
        - **Null Hypothesis (H0): $\rho_{SP} = 0$**.
        - Uses **t-distribution** with **df = n - v**. **$t = r_{SP} * \sqrt{((n - v) / (1 - r_{SP}^2))}$**.
    - **Alternative Understanding**: Pearson's correlation between a variable (A) and the **error (residual)** of another variable (B) after the influence of C has been removed from B (e.g., correlation between anxiety and the residuals of academic achievement explained by intelligence).
- **3.5 Multiple Correlation Coefficient (R)**:
    - Correlation of **one variable (criterion)** with a **linear combination of multiple other variables (predictors)**. Represented as RA.BCD...k.
    - Indicates the proportion of variance in the criterion variable explained by the predictors.
    - **R^2 (Coefficient of Multiple Determination)**: The percentage of variance in the dependent variable explained by the set of predictors.
    - **Formula (for two predictors B, C for criterion A)**: **$R^2_{A.BC} = (r^2_{AB} + r^2_{AC} - 2r_{AB} r_{AC} r_{BC}) / (1 - r_{BC}^2)$**.
    - **R^2 is a biased estimator of population P^2**; needs adjustment for small samples.
    - **Adjusted R^2 (R^2_{adj})**: **$R^2_{adj} = 1 - [(1 - R^2) * (n - 1) / (n - k - 1)]$** (k=number of predictors).
    - **Significance Testing of R (or R^2)**:
        - **Null Hypothesis (H0): $P^2 = 0$** (population R^2 is zero).
        - Uses **F-distribution**: **$F = [R^2 / k] / [(1 - R^2) / (n - k - 1)]$**.
        - **df for F-test**: Numerator df = k; Denominator df = n - k - 1.
        - A significant F-value means the overall model (set of predictors) is significant in explaining variance in the criterion.

Unit 4: Bivariate and Multiple Regression

- **4.0 Introduction**
    - **Regression**: Statistical methods used for **prediction** of one variable from another (or multiple others).
- **4.2 Bivariate and Multiple Regression**

- **Bivariate (Simple Linear) Regression**: Predicts one variable (Y) from a single other variable (X).
  - **Y = Dependent/Criterion Variable** (to be predicted).
  - **X = Independent/Predictor Variable** (used for prediction).
  - "X is regressed on Y" means X predicts Y.
- **4.2.2 Plotting the Relationship**:
  - **Scatter Plots** show relationship.
  - **Perfect Relationship**: All data points fall on a straight line (e.g., hours practice vs. time taken).
  - **Imperfect Relationship**: Data points do not all fall on a straight line (real data); **Linear Regression** finds the "best fit" line.
- **4.2.3 Mean, Variance, and Covariance**: Building blocks (as in Block 2, Unit 1).
- **4.2.4 The Regression Equation**:
  - Population: **$Y = \alpha + \beta X + \varepsilon$**.
  - Sample: **$\hat{Y} = a + bX$** (predicted Y) or **$Y = a + bX + e$** (actual Y with error).
    - **a (Y-intercept)**: Value of Y when X is zero. Interpretation valid only if X values near zero exist.
    - **b (Slope/Regression Coefficient)**: Change in Y for a one-unit change in X. Represents the rate of change.
    - **e (Error/Residual)**: Difference between actual Y and predicted Y ($Y - \hat{Y}$).
- **4.2.5 Ordinary Least Squares (OLS)**:
  - Method to find the **"best fit" regression line** by **minimizing the sum of squared errors (Σe^2 or $\Sigma(Y - \hat{Y})^2$) **.
  - **Calculation of 'b'**: **$b = CovXY / S^2X$**.
  - **Calculation of 'a'**: **$a = Y - bX$**.
  - Total Variance in Y (SST) = Variance Explained by Regression (SSRegression) + Residual Variance (SSResidual). **SST = SSRegression + SSResidual**.
- **4.2.6 Significance Testing of 'b'**:
  - Tests the **null hypothesis that the population slope (β) is zero** (i.e., X does not linearly predict Y).
  - Uses **F-distribution**: **F = SSRegression / SSResidual (Mean Square Regression / Mean Square Residual)**.
  - If F-value is statistically significant, reject H0 (X predicts Y).
- **4.2.7 Accuracy of Prediction**:
  - **Standard Error of Estimate (SY.X)**: Estimate of error in prediction; **$SY.X = \sqrt{(SSResidual / (n - k - 1))}$**.
  - **Percentage of Variance Explained (r^2)**: **$r^2 = SSRegression / SSTotal$**. Indicates how much variance in Y is explained by X.
  - **Proportional Improvement in Prediction (PIP)**: **PIP = $1 - (1 - r^2)$ **.

- **4.2.8 Assumptions Underlying Regression**
    - **Independence** among pairs of scores.
    - **Constant variance of error terms (homoscedasticity)** for each X value.
    - **Linear relationship** between X and Y.
    - Error terms follow **normal distribution** with mean zero and variance one.
    - **Independence of Error Terms** (uncorrelated).
    - Population of X and Y follow **normal bivariate distribution**.
- **4.2.9 Interpretation of Regression**:
    - **Intercept (a)**: Value of Y when X is zero.
    - **Slope (b)**: Change in Y for a one-unit change in X.
    - **r^2**: Percentage of variance in Y explained by X.
    - **F-ratio**: Tests if the overall regression model is significant (if $\beta=0$).
- **4.3 Standardised Regression Analysis and Standardised Coefficients**:
    - Performed using **Z-scores** for both X and Y (ZX and ZY).
    - **Y-intercept (a) becomes zero**.
    - The **standardized slope (b) is equal to the correlation coefficient (r)**.
- **4.4 Multiple Regression**:
    - Predicts a dependent variable (Y) from **multiple predictor variables (X1, X2, ... Xk)**.
    - Equation: **$\hat{Y} = a + b1X1 + b2X2 + ... + bkXk$**.
    - **Multiple R^2**: Total variance in Y explained by the set of predictors.
    - **Adjusted R^2**: An unbiased estimate of population R^2.
    - **Overall Model Significance**: Tested using an **F-test** (ANOVA summary table). A significant F indicates the predictors together significantly explain variance.
    - **Individual Predictor Significance**: Each 'b' can also be tested using a t-test to see its unique contribution.
    - **Methods**: Can be solved **directly (all predictors simultaneously)** or **hierarchically (predictors entered one after another)**.
    - Very useful technique in psychological research for predicting behavior from various factors (e.g., academic achievement from intelligence, work habit, SES).

Here are comprehensive notes for your exam, organized by Block, Unit, and Topic, drawing directly from the provided sources.

Block 3: Inferential Statistics - Normal Distribution, Mean Differences, ANOVA

Unit 1: Characteristics of Normal Distribution

- **1.0 Introduction**
    - Descriptive statistics organize and describe score distributions using histograms, frequency polygons, measures of central tendency, and measures of variability.
    - The **Normal Distribution** and **Normal Curve** (bell-shaped) are helpful for describing individual scores, categorizing groups, and understanding score ranges.
- **1.2 Normal Distribution/Normal Probability Curve**
    - **Concept of Normal Distribution**: In a hypothetical frequency distribution, maximum frequencies concentrate near a central value and gradually taper off symmetrically on both sides.
    - **Concept of Normal Curve**: A frequency polygon of such data resembles a symmetrical "Bell" shape. In a Normal Distribution, **Mean, Median, and Mode are approximately the same**. Many variables in physical, biological, and behavioral sciences (e.g., height, weight, intelligence, achievement) are naturally normally distributed. The Normal Probability Curve (NPC) serves as a **reference curve for mental measurement**, with σ (Sigma) as the unit of measurement.
    - **Theoretical Base**: The NPC is based on the **Law of Probability**, developed by Abraham Demoiver in the 18th century, illustrating the probable occurrence of certain events.
- **1.2.4 Characteristics or Properties of Normal Probability Curve (NPC)**
    - **Symmetrical**: It is symmetrical around its vertical axis (ordinate), meaning left and right halves are mirror images.
    - **Unimodal**: It has only one maximum point (one mode).
    - **Maximum Ordinate at Center**: The highest point is at the midpoint (mean, median, mode). In a unit normal curve, it's 0.3989.
    - **Asymptotic to the X-Axis**: The curve approaches the horizontal axis but never touches it, extending from minus infinity (-∞) to plus infinity (+∞).
    - **Height Declines Symmetrically**: Height decreases symmetrically in both directions from the peak.
    - **Points of Inflection**: Occur at **±1 Standard Deviation (±1σ)** from the mean, where the curve changes from convex to concave.
    - **Area within ±1σ**: Approximately **68.26%** of the curve's area lies within ±1 standard deviation from the mean.
    - **Total Area = 100% Probability**: The total area under the curve is considered to approach 100% probability. For convenience, it's often taken as 10,000 for calculation.

    - **Bilateral**: 50% of the area is to the left of the central ordinate, and 50% is to the right.
    - **Mathematical Model in Behavioral Sciences**: Used as a measurement scale, with ±1σ as the unit. Practically, the curve is considered to end at **±3σ**, encompassing 99.73% of the distribution, with the remaining 0.27% being negligible.
- **1.3 Interpretation of Normal Curve/Normal Distribution**

    - Indicates that the measured trait is normally distributed in the universe.
    - **Most cases (approx. 68.26%) are average** in the trait.
    - Approximately **15.87% of cases are high** in the trait.
    - Similarly, **15.87% of cases are low** in the trait.
    - Suggests a **good test** with high discrimination power and fairly distributed difficulty levels.
- **1.4 Importance of Normal Distribution**

    - Provides a **good fit for many variable facts** in biological, anthropometrical, social, economic, and psychological measurements (e.g., Intelligence, Achievement).
    - Valuable in **educational evaluation and research**. It's a mathematical model, and actual test scores approach it as a limit.
- **1.5 Applications/Uses of Normal Distribution Curve**

    - Determining percentage of cases within given score limits.
    - Determining percentage of cases above or below a score.
    - Determining score limits for a given percentage of cases.
    - Determining percentile rank of an individual or percentile value of a rank.
    - Dividing a group into sub-groups (e.g., assigning grades).
    - Comparing two distributions in terms of overlapping.
    - Determining the relative difficulty of test items.
- **1.6 Table of Areas Under the Normal Probability Curve**

    - To use applications, one must consult the Table of areas.
    - Raw scores (X) are converted to **sigma scores (standard scores or z-scores)** using the formula: **$z = (X - M) / \sigma$**.
    - The table gives fractional parts of the total area (taken as 10,000 for convenience) between the mean and various σ distances.
    - **Points to Keep in Mind**:
        - Convert scores to z-scores.
        - Mean is the reference point (zero).
        - Area in proportion can be converted to percentage.
        - Use **absolute values of z**; negative z indicates score/area below the mean, positive z indicates above the mean.

- **1.9 Divergence in Normality (The Non-Normal Distribution)**
    - Occurs when the curve lacks symmetry.
    - **Skewness**: When mean and median fall at different points, shifting the center of gravity.
        - **Normal Distribution**: Skewness is **zero (SK = 0)**.
        - **Negative Skewness (Skewed to the Left)**: Scores are massed at the high end (right side), spreading gradually towards the low end (left side). **Median is higher than the Mean**.
        - **Positive Skewness (Skewed to the Right)**: Scores are massed at the low end (left side), spreading gradually towards the high end (right side).
    - **Kurtosis**: Refers to the **peakness** of the curve.
        - **Leptokurtosis**: The curve is **more peaked and narrow** than the normal curve, with scores shrinking towards the center.
        - **Platykurtosis**: The curve has a **flatter peak** than the normal curve.
        - **Normal Distribution**: Kurtosis is **0.263 (KU = 0.263)**. KU > 0.263 is platykurtic, KU < 0.263 is leptokurtic.
- **1.10 Factors Causing Divergence**
    - **Selection of Sample**: Small or biased samples can cause skewness; small, homogeneous groups yield leptokurtic distributions; small, heterogeneous groups yield platykurtic distributions.
    - **Unsuitable or Poorly Made Tests**: Too easy (scores pile high) or too hard (scores pile low) tests.
    - **Non-Normal Trait**: Real lack of normality in the measured trait (e.g., interest, attitude).
    - **Errors in Test Construction/Administration**: Unstandardized tests, poor item analysis, unclear instructions, timing errors, scoring errors, motivation.
- **1.11 Measuring Divergence**
    - Important for test construction and sample representativeness.
    - **Measuring Skewness**:
        - **Observation Method**: Looking at the longer tail of the frequency polygon (right = positive, left = negative).
        - **Statistical Method**:
            - **SK = 3 * (Mean - Median) / σ**
            - SK = (P90 - P50) - (P50 - P10) / (P90 - P10) / 2
            - Normal curve has SK = 0.
    - **Measuring Kurtosis**:
        - **Observation Method**: Inspecting peak thinness (leptokurtic) or flatness (platykurtic).
        - **Statistical Method**: **KU = Q / (P90 - P10)** (where Q = quartile deviation).
        - Normal distribution has KU = 0.263.

Unit 2: Significance of Mean Differences, Standard Error of the Mean

- **2.0 Introduction**
    - Inferential statistics draws inferences about a **population** based on a **sample**.
    - Crucial to determine the **reliability** or **trustworthiness** of sample measures before generalizing.
    - This unit focuses on **Standard Error of a sample mean** and comparing two sample means.
    - A **"significant difference"** is larger than expected by chance or sampling fluctuations.
- **2.2 The Concept of Parameters and Statistics**
    - **Estimate (Statistic)**: Statistical measurements (mean, SD, correlation) obtained from a **sample**. Represented by English alphabets (M, S.D., r).
    - **Parameter (True Measure)**: Statistical measurements obtained from the **entire population**. Represented by Greek letters (μ, σ, η) or bar notation ($\bar{M}$, $S.\bar{D}.$, $\bar{r}$).
    - **Goal**: To forecast population parameters from sample statistics with known accuracy.
- **2.3 Significance and Level of Significance**
    - **Significance/Trustworthiness**: The degree to which a sample mean (M) represents its population parameter (Mpop).
    - A sample mean is statistically trustworthy if the difference between Mpop and M could have arisen "by chance" with low probability.
    - **Level of Significance (or Confidence)**: The confidence level to which a forecast is made.
        - Commonly used in psychology: **99% (0.01 level)** and **95% (0.05 level)**.
        - **0.01 level**: In 100 trials, only 1 occasion will the sample mean fall outside Mpop ± 2.58 S.E.. This is more rigorous.
        - **0.05 level**: In 100 trials, only 5 occasions will the sample mean fall outside Mpop ± 1.96 S.E..
        - Values 1.96 and 2.58 are from the 't' distribution table.
- **2.4 Sampling Error and Standard Error**
    - Individual scores have **measurement error** (due to test reliability, personal observation) and **statistics error**.
    - **Statistics Error**: Errors in sample statistical measurements.
        - **Sampling Error**: Difference between the entire population mean and the sample mean (Mpop ~ M). Lower difference implies better representation.
        - **Standard Error (S.E.M)**: **Standard deviation of the sampling distribution of statistics** (e.g., sample means). It measures the

intra-differences in sample measurements from multiple samples taken from the same population.
        - **Formula**: **S.E.M or σM = σ / √N**
            - σ: Population standard deviation (often replaced by sample SD, σ, in practice).
            - N: Sample size.
        - **S.E.M depends on**: (i) variability (σ) of population scores and (ii) sample size (N).
        - Low S.E.M occurs with homogeneous population (low σ) and large sample size (large N), indicating a reliable and representative sample.
- **2.5 ‘t’ Ratio and ‘t’ Ratio Distribution Table**
    - **Sampling Distribution**: A frequency distribution of sample statistics (e.g., 100 means from 100 samples), which is typically "Bell-Shaped" (normal curve).
    - **'t' Ratio**: A **standard score** for sample measures, analogous to z-score.
        - **Formula**: **t = (M - Mpop) / S.E.M**
            - M: Sample mean, Mpop: Population mean, S.E.M: Standard Error of Mean.
        - Indicates how far the sample mean deviates from the population mean.
    - **Student's 't' Distribution**: The sampling distribution of 't' ratios. Its mean is zero, and its standard deviation is ±σt.
        - **Affected by sample size**: Becomes more leptokurtic as sample size decreases. Approaches normal distribution as N becomes large.
        - Fisher's 't' table provides 't' values for different N (or degrees of freedom) and significance levels.
- **2.6 Standard Error of Sample Statistics – The Sample Mean**
    - **Meaning**: Measures how the mean is affected by measurement errors and sampling fluctuations.
    - **Large Sample (N > 30)**:
        - Formula: **S.E.M = σ / √N** (if population SD known).
        - **In practice**: **S.E.M = σ / √N** (using sample SD, σ).
        - **Example**: For N=225, M=40, σ=12, S.E.M = 0.80. The sample mean is trustworthy at 95% probability.
    - **Degree of Freedom (df)**:
        - "Degrees of Freedom to Vary" refers to the number of values that are free to vary after restrictions are imposed.
        - For calculating Mean or SD, df is **N-1** (one restriction due to fixed sum). For correlation, it's N-2.
        - For large samples, N or N-1 doesn't significantly change S.E.M, but for small samples, N-1 is crucial.
        - As df approaches 500+, 't' values for 95% and 99% become constant (1.96 and 2.58), approaching Z-distribution.

    - **Small Sample (N ≤ 30)**:
        - Formula: **S.E.M = σ / √(N-1)** (using sample SD, σ, as population σ is unknown).
        - **Example**: For N=17, M=58, σ=8, S.E.M = 2.0. 't' for 16 df at .01 level is 2.92. Calculated 't' (2.0) is less, so the mean is trustworthy at 99% confidence.
- **2.7 Application of the Standard Error of Mean**
    - **Estimation of Population Statistics (Mpop)**:
        - Estimation is always a **range (Fiduciary limits)**, not a point.
        - **Formulas**:
            - Mpop = M ± **2.58 σM** (at 0.01 level)
            - Mpop = M ± **1.96 σM** (at 0.05 level)
        - **Example**: N=400, M=56, σ=14. At 99% confidence, Mpop = 56 ± 2.58 * 0.70 = **54.194 to 57.806** (approx. 54-58).
    - **Determination of Sample Size (N)**:
        - Used to estimate required sample size for specific S.E.M.
        - **Example**: σ=20, S.E.M=2. N = (σ / S.E.M)^2 = (20/2)^2 = **100**.
        - **Example (with confidence level)**: σ=16, max S.E.M=1.90, at 99% (Z=2.58). N = (2.58 * σ / S.E.M)^2 = (2.58 * 16 / 1.90)^2 = **472**.
- **2.8 Importance and Application of Standard Error of Mean**
    - Helps in **drawing concrete conclusions** in research.
    - Determines **reliability of a sample** and its statistics.
    - **Estimates population parameters** within a confidence interval, making research feasible and economical when population is unknown.
    - Determines **sample size** for studies.
    - Assesses **significance of difference between two groups** by eliminating chance errors.
- **2.9 The Significance of the Difference Between Two Means**
    - Purpose: To determine if an observed difference between two sample means is **statistically significant** (real and dependable) or due to **chance**.
    - Depends on: (i) **Standard Error of Difference between the two means (S.E.DM)** and (ii) **Levels of significance**.
    - **S.E.DM (Standard Error of the Difference of Two Means)**: Standard deviation of the sampling distribution of mean differences.
        - **Formula (Independent Populations)**: **S.E.DM = √(σ1^2/N1 + σ2^2/N2)**
            - σ1, σ2: Standard deviations of sample 1 & 2. N1, N2: Sizes of sample 1 & 2.
    - **Critical Ratio (C.R.)**: The Z-score for the difference of two sample means.
        - **Formula**: **C.R. = (M1 ~ M2) / S.E.DM** (assuming population mean difference is zero).
    - **Levels of Significance (for Mean Differences)**:

        - **Z = 1.96 (0.05 level)**: If CR ≤ 1.96, difference is within acceptance area, null hypothesis retained. If CR > 1.96, difference is in rejection area, null hypothesis rejected.
        - **Z = 2.58 (0.01 level)**: If CR ≤ 2.58, difference is significant at .01 level. If CR > 2.58, null hypothesis rejected at .01 level.
    - **Null Hypothesis (Ho)**: States there is **no true difference** between two population means; observed differences are by chance.
        - **Alternative Hypothesis (HA)**: Accepted if Ho is rejected.
    - **Basic Assumptions for Testing Significance**:
        - Variable is normally distributed in the universe.
        - No difference in population means (M1=M2 assumed under Ho).
        - Samples are randomly drawn.
    - **Two-Tailed vs. One-Tailed Test**:
        - **Two-tailed**: Used when interested if two groups differ (M1-M2=0) without specific direction. Considers both positive and negative deviations.
        - **One-tailed**: Used when concerned with the **direction** of difference (e.g., only positive effect/gain). Uses only one side of the normal curve. Z values change (e.g., 1.64 for 95% one-tailed vs. 1.96 two-tailed).
    - **Uncorrelated (Independent) vs. Correlated (Dependent) Sample Means**:
        - **Independent**: Samples from different populations or uncorrelated tests on the same sample.
        - **Correlated**: Single group tested in two situations (pre/post) or matched pairs. Requires modified S.E.DM formula.
- **2.10 Significance of Two Large Independent or Uncorrelated Sample Means**
    - Formula: **CR = (M1 ~ M2) / √(σ1^2/N1 + σ2^2/N2)**.
    - **Example**: Boys M=52, SD=13, N=65; Girls M=48, SD=12, N=60. CR = 1.79. df = (65-1)+(60-1) = 123. Table 't' at .05 for df 125 is 1.98. Since 1.79 < 1.98, **null hypothesis is accepted**. No significant difference in intelligence between boys and girls.
- **2.11 Significance of the Two Small Independent or Uncorrelated Sample Means**
    - Formula (when raw data available): **t = (M1 ~ M2) / √[(Σd1^2 + Σd2^2) / (N1+N2-2) * (1/N1 + 1/N2)]**.
    - Formula (when only statistics given): **t = (M1 ~ M2) / √[((N1-1)σ1^2 + (N2-1)σ2^2) / (N1+N2-2) * (N1+N2)/(N1*N2)]**.
    - **Example**: Rural vs. Urban boys' attitude (small sample data). Calculated t = 4.46. df = 13. Table 't' for 13 df at .05 is 2.16. Since 4.46 > 2.16, **null hypothesis is rejected**. Significant difference in attitude.
- **2.12 Significance of the Two Large Correlated Samples**
    - Formula: **t = (M1 ~ M2) / √(σM1^2 + σM2^2 - 2 * r12 * σM1 * σM2)**
        - σM1 = σ1/√N, σM2 = σ2/√N. r12 = correlation between scores.

    - **Example**: Intelligence test pre/post on N=400. M1=25, SD1=8; M2=30, SD2=5; r=0.65. Calculated t = 1.95. df = 399. Table 't' at .01 for 399 df is 2.59. Since 1.95 < 2.59, **null hypothesis is retained**. No significant difference.
    - **Example (One-tailed test for gain)**: Achievement test pre/post on N=100. M1=52.5, SD1=7.25; M2=58.7, SD2=5.3; r=0.50. Calculated t = 9.54. df = 99. Table 't' at .02 for 99 df is 2.36 (one-tailed). Since 9.54 > 2.36, **null hypothesis is rejected**. Gain is highly significant, training effective.
- **2.13 Significance of Two Small Correlated Means**
    - **Direct Method**: Formula: **t = (M1 ~ M2) / √(Sm1^2 + Sm2^2 - 2 * r12 * Sm1 * Sm2)** where Sm is standard error of small sample mean.
    - **Difference Method**: Calculate difference (D) for each pair. Calculate Mean of Differences (MD) and Standard Deviation of Differences (σD).
        - **t = MD / SEDM** where SEDM = σD / √N.
    - **Example**: Pre-test/Post-test scores for 12 subjects. Calculated t = 4.88. df = 11. Table 't' at .02 for 11 df is 2.72. Since 4.88 > 2.72, **null hypothesis is rejected**. Gain is significant.
- **2.14 Points to be Remembered While Testing Significance**
    - Set null (Ho) and alternative (H1) hypotheses.
    - Decide significance level (.05 or .01).
    - Decide one-tailed or two-tailed.
    - Determine if samples are large or small.
    - Calculate CR or 't' ratio.
    - Calculate degrees of freedom (df): N1+N2-2 for independent; N-1 for correlated.
    - Consult 't' distribution table.
    - Compare calculated 't' with table 't'.
    - Interpret results: Reject Ho (significant difference) or Accept Ho (no significant difference, due to chance).
- **2.15 Errors in Interpretation**
    - **Type-I Error (α Error)**: Rejecting the null hypothesis when it is true.
    - **Type-II Error (β Error)**: Accepting the null hypothesis when it is false.

Unit 3: One Way Analysis of Variance

- **3.0 Introduction**
    - **Limitation of t-test**: Only compares two means.
    - **Problem**: If comparing more than two groups (e.g., A, B, C), applying multiple t-tests (A-B, B-C, A-C) becomes cumbersome and increases the chance of **Type I error**.
    - T-test only considers differences in means, not in variance or homogeneity.

    - **Solution**: **Analysis of Variance (ANOVA)** and the **F-test** are more suitable for comparing more than two groups, focusing on **variance**.
- **3.2 Analysis of Variance (ANOVA)**

    - **Definition**: A group of inferential statistical procedures that **divides observed variation into different parts**, each assigned to a known source or factor.
    - Used to test the **significance of differences between means of multiple populations**.
    - **F-ratio**: **F = (Variance between groups) / (Variance within groups)**.
    - Developed by **Sir Ronald Fisher**.
    - **Meaning of Variance**: The **average of sum of squares of deviation from the mean** of a distribution. It's the square of Standard Deviation (SD^2).
    - **Characteristics of Variance**:
        1. Measures variability: **among groups** and **within group** differences.
        2. Always **positive**.
        3. Like an **area** (unlike SD, which has direction).
        4. Remains the same when a constant is added or subtracted from data.
- **3.2.3 The Procedure of Analysis of Variance (ANOVA)**

    - Used when comparing two or more groups on a dependent variable, or studying different treatments of a single independent variable.
    - **One-way ANOVA**: Involves one independent variable (factor).
    - **Steps** (General outline):
        1. **Correction Term (Cx)**: Sum of all scores, squared, divided by total N.
        2. **Sum of Squares Total (SST)**: ΣX^2 - Cx.
        3. **Sum of Squares Among Groups (SSA)**: Σ(Group Sum)^2 / n - Cx.
        4. **Sum of Squares Within Groups (SSW)**: SST - SSA.
        5. **Mean Sum of Squares Among Groups (MSSA)**: SSA / (k-1) (where k = number of groups).
        6. **Mean Sum of Squares Within Groups (MSSW)**: SSW / (N-k) (N = total units).
        7. **F Ratio**: MSSA / MSSW.
        8. **Summary Table of ANOVA**: | Source of Variance | df | S.S. | M.S.S. | F Ratio | | :---------------- | :---- | :--- | :----- | :------ | | Among Groups | k-1 | SSA | MSSA | MSSA/MSSW | | Within Groups | N-K | SSW | MSSW | | | Total | N-1 | | | |
        9. **Evaluate F-ratio**: Compare calculated F with table F values (using df1 = k-1 and df2 = N-k) at .05 and .01 levels.
            - **Insignificant F**: If calculated F < table F, **null hypothesis is retained**. No significant difference among means. Stop.
            - **Significant F**: If calculated F > table F, **null hypothesis is rejected**. Significant difference exists. Proceed to **post-hoc tests** (e.g., t-tests) to find which means differ.

- **3.2.5 Assumptions Underlying Analysis of Variance**
    - **Normality**: Dependent variable normally distributed in the population.
    - **Homogeneity of Variance**: Variances in different sets of scores do not differ beyond chance (σ1=σ2=σ3...=σk).
    - **Random Samples**: Samples are randomly selected.
    - **No significant difference in means of various samples** (This is the null hypothesis, which ANOVA tests).
- **3.2.6 Relationship between 'F' test and 't' test**
    - **Complementary**: 't' follows 'F' if 'F' is significant (for specific comparisons). 'F' is used if 't' is not significant (as 't' doesn't evaluate within-group variance).
    - **Mathematical relation**: **F = t^2** or **t = √F**.
- **3.2.7 Merits or Advantages of Analysis of Variance**
    - Improved over 't'-test: Evaluates both 'between' and 'within' group variance.
    - Ascertains differences among several groups/treatments simultaneously (economical).
    - Can involve more than one variable (main and interaction effects).
    - Basis for experimental designs (e.g., simple random design).
    - Essential if 't' is not significant to analyze differences.
- **3.2.8 Demerits or Limitations of Analysis of Variance**
    - Relies on assumptions (normality, homogeneity); violation can invalidate inferences.
    - 'F' value provides global findings, but **cannot specify which means differ** (requires 't'-test post-hoc).
    - Time-consuming and requires skill for calculations and interpretations.
    - Requires an F-table.
- **3.3 F Ratio Table and its Procedure to Use**
    - F-table used to evaluate calculated F-ratio.
    - Requires **two degrees of freedom**: **df1 (numerator, larger variance)**, typically (k-1); and **df2 (denominator, smaller variance/error variance)**, typically (N-k).
    - Table provides F-values at .05 (normal print) and .01 (bold/dark print) levels.
    - If calculated F is less than table F, it's not significant; if greater, it's significant.

Unit 4: Two Way Analysis of Variance

- **4.0 Introduction**
    - **One-way ANOVA**: Studies the effect of **one independent variable** on a dependent variable.

- **Two-way ANOVA**: Used to study the **effect of two independent variables** on a single dependent variable.
- Allows studying: (1) independent effect of Drug A, (2) independent effect of Drug B, and (3) **joint or interactional effect (A x B)**.
- Leads to **factorial designs** (e.g., 2x2 design for family type x gender effects on cognitive development).

- **4.2 Two Way Analysis of Variance**

  - Simultaneously considers two independent variables.
  - Involves analysis **column-wise (vertically)** and **row-wise (horizontally)**.
  - Tests three types of effects:
    - **Main Effect of Variable A** (e.g., SES on IQ)
    - **Main Effect of Variable B** (e.g., Gender on IQ)
    - **Interaction Effect (A x B)** (e.g., SES x Gender on IQ)
  - **Summary Table of Two-way ANOVA**: | Source of Variance | df | SS | MSS | F Ratio | | :---------- | :---------- | :------- | :------- | :---------- | | Between Group A | k1-1 | SSA | MSSA | F1 (MSSA/MSSW) | | Between Group B | k2-1 | SSB | MSSB | F2 (MSSB/MSSW) | | Interaction A x B | (k1-1)(k2-1) | SSAxB | MSSAxB | F3 (MSSAxB/MSSW) | | Within Groups | N-k | SSW | MSSW | | | Total | N-1 | | | |
  - **Interpretation of F-ratios**:
    - All F-ratios insignificant: No independent or interaction effect. Null hypothesis retained.
    - All F-ratios significant: Significant independent and interaction effects. Null hypothesis rejected. Requires post-hoc tests if variables have >2 levels.
    - Some F-ratios significant: Partial retention/rejection. Requires post-hoc tests if variables have >2 levels.

- **4.3 Interactional Effect**

  - The **combined or joint effect** of two or more independent variables on the dependent variable.
  - Crucial for understanding how variables work together.
  - Individual effects might be significant, but their combination can lead to very high, very low, or nullified effects on the dependent variable.
  - **Graphical Representation**:
    - **No Interaction Effect**: Lines representing the levels of one factor across the levels of another factor will be approximately **parallel**.
    - **Significant Interaction Effect**: Lines will **intersect or show non-parallel patterns**.

- **4.4 Merits and Demerits of Two Way ANOVA**

  - **Merits/Advantages**:
    - Analyzes **main effects** and **interaction effects**.
    - Analyzes effects of **more than two factors**.

        - Used for **factorial designs** and complex experimental studies.
    - **Demerits/Limitations**:
        - **Global findings for main effects**: Cannot specify exact differences (requires 't'-test post-hoc if >2 levels).
        - **Assumptions**: Relies on normality and homogeneity of variance; violation can lead to spurious results.
        - **Complex and time-consuming**.
        - Complexity increases with more factors, making interpretation difficult.
        - Requires high arithmetical, calculative, imaginative, and logical ability.

Block 4: Non-Parametric Statistics

Unit 1: Rationale for Non-Parametric Statistics

- **1.0 Introduction**
    - Human behavior is unpredictable and hard to quantify.
    - **Parametric statistics** require **normal distribution assumptions**.
    - **Non-parametric statistics** do **not require these assumptions** and are not necessarily compared with the normal curve.
- **1.2 Definition of Non-parametric Statistics**
    - Techniques that **do not rely on data belonging to any particular distribution**.
    - Includes:
        1. **Distribution-free methods**: Interpretation does not depend on parametrized distributions.
        2. **Non-structural models**: Model grows in size to accommodate data complexity.
    - Deals with statistics based on **ranks of observations**, not necessarily interval or ratio scores.
    - Applicable when data has ranking but no clear numerical interpretation (e.g., preferences, ordinal scales).
    - **Wider applicability** due to fewer assumptions. Simplicity and robustness reduce improper use.
    - **Statistic**: Characteristic of a sample (e.g., mean).
    - **Parameter**: Characteristic of a population (e.g., population average).
    - Distinction between parametric and non-parametric tests is primarily based on **level of measurement**.
- **1.3 Assumptions of Parametric and Non-parametric Statistics**
    - **Parametric Test Assumptions**:
        1. **Normal distribution** of the dependent variable.

2. Minimum of **Interval data** level of measurement.
3. **Adequate sample size** (>30 recommended per group).
4. Independence of observations (except paired data).
5. Randomly drawn observations for dependent variable.
6. **Equal variance** among sample populations.
7. Hypotheses usually about **numerical values (mean)**.

- **Non-parametric Test Assumptions**:
  1. Independence of observations (except paired data).
  2. Continuity of variable under study.
  3. **Fewer assumptions regarding population distribution**.
  4. Sample sizes are **less stringent**.
  5. Measurement level can be **nominal or ordinal**.
  6. Primary focus on **rank ordering or frequencies**.
  7. Hypotheses about **ranks, medians, or frequencies**.
- **Violations**: Level of measurement, sample size, and normal distribution are often violated in psychology research.

- **1.3.1 Level of Measurement**

  - Parametric tests typically require **interval or ratio level**.
  - Non-parametric tests can be used with **all levels**, most frequently with **nominal and ordinal data**.
  - **Nominal Data (Categorical)**: Mutually exclusive, named categories with **no implied ordering** (e.g., yes/no, male/female). Data are counted frequencies. Cannot obtain normal distribution.
  - **Ordinal Data (Rank-order)**: Quantitative **order** in mutually exclusive categories, but **no indication of value differences between positions** (e.g., pain scales, army ranks).
  - **Interval Data**: Ordered, mutually exclusive categories with **equidistant divisions**. Zero does not represent absence of value.
  - **Ratio Data**: Highest level. Equal intervals and a **meaningful zero point** (e.g., weight, blood pressure).
- **1.3.5 Sample Size**

  - Parametric tests assume **adequate (large) sample size**. This assumption is often violated in health science research due to small convenience samples or limited funding.
  - Small sample sizes reduce statistical power and increase Type II error risk.
  - Non-parametric techniques can sometimes **gain adequate power with small samples**.
  - **No consensus on "very small" sample size**.
  - Decision on parametric vs. non-parametric depends on data nature, sample size, measurement level, population distribution knowledge, and distribution shape.
- **1.3.6 Normality of the Data**

- Parametric tests assume **normal distribution of dependent variable**.
- In real research, population characteristics are often unknown. Sample sizes may be too small for assumptions.
- Non-parametric statistics are for when **nothing is known about the variable's distribution**.
- Many variables in social/health sciences are **non-normally distributed** (e.g., rare diseases).
- Empirical studies show widespread **non-normal distributions** in achievement and psychometric measures.

- **1.4 The Use of Non-parametric Tests**

  - A non-parametric test is available for each main parametric technique.
  - Categories of use:
    1. **Differences between Independent Groups**: Non-parametric alternatives to t-test (independent samples) are **Wald-Wolfowitz runs test, Mann-Whitney U test, Kolmogorov-Smirnov two-sample test**. For multiple groups (ANOVA equivalent): **Kruskal-Wallis analysis of ranks, Median test**.
    2. **Differences between Dependent Groups**: Non-parametric alternatives to t-test (dependent samples) are **Sign test, Wilcoxon's matched pairs test**. For dichotomous variables: McNemar's Chi-square test. For repeated measures ANOVA equivalent: **Friedman's two-way analysis of variance, Cochran Q test**.
    3. **Relationships between Variables**: Non-parametric equivalents to Pearson 'r' are **Spearman R, Kendall Tau**. For categorical variables: **Chi-square test, Phi coefficient, Fisher exact test**. For inter-rater agreement: Kendall coefficient of concordance.
    4. **Descriptive Statistics**: For non-normally distributed data or rank-order information.

- **1.4.7 Advantages and Disadvantages of Non-parametric Statistics**

  - **Advantages**:
    1. **Less stringent demands/fewer assumptions** of data (e.g., normality not required for Wilcoxon tests). This implies **robustness**.
    2. **Quick answer with little calculation** (e.g., Sign test, Median test).
    3. Provide **objectivity when underlying scale is unreliable** or artificial. Data converted to ranks.
    4. Historically, easy to construct tables of exact critical values (for rank tests).
    5. Applicable when data **don't constitute a random sample** from a larger population (using randomization models).
    6. Wide application in social/health sciences, especially for constructs without quantitative measures and **small sample sizes**.
  - **Disadvantages**:

1. **No parameters to describe**: Difficult to make quantitative statements about actual differences (e.g., Sign test doesn't provide confidence intervals or magnitude of difference).
2. **Throw away information**: Ranks preserve order but discard actual values.
3. **Less powerful**: Cannot be as powerful as parametric counterparts when parametric tests *can* be used (i.e., when assumptions are fully met).

- **1.5 Misconceptions about Non-parametric Tests**
    - **Less powerful**: Often perceived as inferior, but parametric tests are only more powerful if *all* assumptions are met.
    - **Limited application**: Thought to be restricted to simple designs, but non-parametric techniques exist for complex designs (e.g., structural equation modeling). For every parametric test, there's an equally or more appropriate non-parametric equivalent.
    - Lack of use due to **limited exposure** and portrayal as less important in basic statistics courses.

Unit 2: Mann Whitney 'U' Test for Two Sample Test

- **2.2 Definition of Two Sample Tests**
    - Tests of **independence** (not goodness of fit).
    - **Null Hypothesis (Ho)**: Two variables (e.g., X and Y) are independent.
    - Used to determine if two groups differ in performance when **parametric assumptions (normality, sample size > 30) are not met**.
    - Example: Comparing marks in History for males (nominal, dichotomous) and females.
    - Example: Comparing final exam scores for two groups receiving different teaching methods (X1 and X2).
- **2.3 Mann Whitney 'U' Test**
    - **Non-parametric analog of the two-sample t-test for independent samples**.
    - Used when measurement is at least **ordinal**.
    - Analyzes the **degree of separation (overlap)** between two independent groups (e.g., Experimental and Control).
    - **Null Hypothesis (Ho)**: The two sets of scores are **samples from the same population** (do not differ systematically).
    - **Alternative Hypothesis**: Scores differ systematically. Can be directional (one-tailed).
    - **Statistic calculated**: U or U'.
        - U1 = number of Es less than Cs.
        - U2 = number of Cs less than Es.

        - U = smaller of U1, U2. U' = larger of U1, U2.
    - Used to compare **location parameters (medians)** or scale parameters of two populations.
- **2.4 Relevant Background Information on 'U' Test**
    - Employed with **ordinal (rank-order) data**.
    - Significant result indicates a **significant difference between two sample medians**, suggesting populations with different median values.
    - Can be used when data are already in rank order, or when interval/ratio data are **transformed into ranks** due to violation of parametric assumptions (e.g., normality, homogeneity of variance).
    - **Information is sacrificed** when transforming to ranks, which is a reluctance for some researchers.
    - **Assumptions**:
        - Randomly selected samples.
        - **Two samples are independent**.
        - Original variable is continuous (though often used with discrete variables).
        - **Underlying distributions are identical in shape** (implies equal dispersion/homogeneity of variance), though not necessarily normal.
    - Less affected by violation of homogeneity of variance than t-test.
    - Ranking can **reduce or eliminate the impact of outliers**.
- **2.5 Step by Step Procedure for 'U' Test for Small Sample (N ≤ 20 per set)**
    - **Rank all data together** (both groups combined), from lowest (Rank 1) to highest.
    - Find the **sum of ranks** for the smaller sample (ΣR1).
    - Find the **sum of ranks** for the larger sample (ΣR2).
    - **Calculate U**:
        - **$U = N1N2 + [N1(N1 + 1) / 2] - \Sigma R1$**
        - **$U' = N1N2 + [N2(N2 + 1) / 2] - \Sigma R2$**
        - Take the smaller of U and U' for significance testing.
    - **Determine Significance**: Look up the smaller U value in Table H for N1 and N2.
        - If **observed U is smaller than or equal to the table value**, the result is **significant**.
    - Translate results back to the experiment.
    - **Example**: Assessment Center Rating by Two Teams. N1=12, N2=13. ΣR1=148, ΣR2=177. U=86, U'=70. Critical U for N1=12, N2=13, α=0.05 (two-tailed) is 41. Since 70 > 41, **null hypothesis is accepted**. No significant difference in ratings.
- **2.6 Step by Step Procedure for 'U' Test for Large Sample (N > 20 per sample)**
    - The sampling distribution of U becomes approximately **normal**.
    - **Perform a z-test**:
        - **$Z = [U - (N1N2) / 2] / \sqrt{[N1N2 (N1 + N2 + 1) / 12]}$**
    - Significance: For two-tailed, z > 1.96; for one-tailed, z > 1.64.
    - Less used for large samples as parametric assumptions become less important.

- **2.8 Wilcoxon Matched Pair Signed Rank Test**
    - **Non-parametric test for 2 repeated (or correlated) measures** when measurement is at least ordinal.
    - Unlike the Sign test, it considers the **magnitude of the difference**.
    - **Procedure**:
        - Obtain **difference scores** between N pairs.
        - **Rank** these difference scores (ignoring sign).
        - Sum ranks of negative differences (T-) and positive differences (T+).
        - **Test statistic T is the smaller of these two sums**.
        - Differences of 0 are eliminated.
    - **Null Hypothesis**: Sum of positive ranks should not differ from sum of negative ranks beyond chance.
- **2.9 Relevant Background Information on Wilcoxon Test**
    - Used when assumptions of **t-test for two dependent samples** are violated.
    - Extension of Wilcoxon signed-ranks test (single sample version).
    - Requires two interval/ratio scores for each subject (or matched pair) under two conditions.
    - **Hypothesis evaluated**: Whether median of difference scores in underlying populations equals zero.
    - **Assumptions**:
        - Randomly selected sample.
        - Original scores are interval/ratio data.
        - Distribution of difference scores is **symmetric about the median**.
    - Unlike other rank tests, it ranks **difference scores**, not original scores.
- **2.10 Step by Step Procedure for Wilcoxon Test for Small Sample (N ≤ 25 pairs)**
    - Obtain **differences** between each pair of readings, considering sign.
    - **Rank order** these differences (ignoring sign), rank 1 to smallest difference.
    - Obtain **T, the sum of ranks for differences with the less frequent sign**.
    - Consult Table J. If **observed T is equal to or less than the table value**, there is a **significant difference**.
    - Translate results to experiment.
    - **Example**: Reaction time after whisky vs. sober for 8 twins. Calculated T = 3 (sum of negative ranks: 1+2). Table T for N=8 is 4. Since 3 < 4, **significant difference**. Reaction time is higher after whisky.
- **2.11 Step by Step Procedure for Wilcoxon Test for Large Sample (N > 20 pairs)**
    - Sampling distribution of T approaches **normal distribution**.
    - **Perform a z-test**:
        - **$Z = \{T - [N(N+1) / 4]\} / \sqrt{[N(N+1)(2N+1) / 24]}$**
    - Significance decisions similar to Mann-Whitney large sample case ($z > 1.96$ for two-tailed, $z > 1.64$ for one-tailed).

- **2.13 Comparison of Mann-Whitney 'U' Test and Wilcoxon MPSR test with t-test**

    - **Power efficiency**: Usually slightly lower than corresponding t-test, but compares very favorably.
    - **Wider application**: Can be used when t-test assumptions (e.g., normality, large differences in variance) obviously do not apply.
    - Non-parametric tests are simpler to compute, have fewer assumptions, and can be used more widely.
    - Recommended strategy: Use **t-test unless data is in ranks, or sample is small and obviously non-normal or has large variance differences**.
    - Using non-parametric statistics is not inappropriate even when t-tests could have been used.

Unit 3: Kruskal Wallis Analysis of Variance

- **3.0 Introduction**

    - Addresses the problem of **comparing more than two groups** (>2 groups).
    - Multiple t-tests for >2 groups lead to **cumbersome calculations** and increased **Type I error**.
    - **Solution**: **Analysis of Variance (ANOVA)**.
- **3.2 Analysis of Variance**

    - **Definition**: Group of inferential statistical procedures by Sir Ronald Fisher, examining variability in a response variable (y).
    - Compares several populations (groups) regarding a quantitative variable.
    - **Null Hypothesis (Ho)**: **Population means are equal** ($\mu1 = \mu2 = \mu3 = \mu4$).
    - **Alternative Hypothesis (H1)**: Population means are not equal (at least two differ).
    - **Logic similar to t-test**: One-way ANOVA gives identical results to t-test for two groups.
    - **Assumptions**:
        - **Random, independent sampling**.
        - **Normal population distributions**.
        - **Equal variances** within the k populations.
    - Assumptions 2 & 3 can be relaxed for large samples, or if sample sizes are roughly equal (for small samples).
- **3.3 Introduction to Kruskal-Wallis ANOVA Test**

    - **Non-parametric analogue to ANOVA**.
    - Compares the **medians** of several (more than two) populations.
    - Viewed as **ANOVA based on rank-transformed data**.
    - **Null Hypothesis (Ho)**: The population medians are equal.

    - **Alternative Hypothesis (H1)**: The population medians differ.
- **3.4 Relevant Background Information on Kruskal Wallis ANOVA Test**
    - Also known as **Kruskal-Wallis one-way analysis of variance by ranks** (developed by Kruskal and Wallis, 1952).
    - Extension of the **Mann-Whitney U test** for >2 independent samples.
    - Significant result: Indicates a significant difference between at least two sample medians in the set.
    - Used when data are in rank-order format, or when interval/ratio data are transformed to ranks due to **violation of parametric ANOVA assumptions** (e.g., normality, homogeneity of variance).
    - **Information is sacrificed** when transforming to ranks.
    - **Assumptions**:
        - **Each sample randomly selected**.
        - **k samples are independent**.
        - Dependent variable is a **continuous random variable** (though often used with discrete).
        - **Underlying distributions are identical in shape** (implies equal dispersion/homogeneity of variance), but **do not have to be normal**.
    - Test statistic less affected by homogeneity of variance violation than F-distribution.
    - Ranking reduces or eliminates outlier impact.
- **3.5 Step by Step Procedure for Kruskal Wallis ANOVA**
    - **Rank all numbers in the entire data set** (all samples combined) from smallest to largest. Use average ranks for ties.
    - **Total the ranks for each sample** (T1, T2, ..., Tk).
    - **Calculate the Kruskal-Wallis test statistic (H)**:
        - **H = [12 / N (N+1)] [Σ((ΣR)^2 / n)] – 3(N + 1)**
            - N = total number of cases, n = cases in a group, ΣR = sum of ranks for a group.
    - **Find the p-value**. For small samples, use Table H.
    - **Conclusion**: Reject Ho if p-value is significant (or calculated H exceeds critical value).
    - **Example**: Errors in test under 3 noise conditions (A, B, C), 4 participants each. H = 5.537. Critical value for k=3, n=4,4,4 at α=0.05 is 5.692. Since 5.537 < 5.692, **accept null hypothesis**. No difference in performance based on noise level.
- **3.6 Considerations for Large Sample**
    - When sample size increases (or Table H is not applicable), use **chi-square table** for critical values, with **degrees of freedom (k – 1)**.
    - Chi-square distribution provides a more accurate estimate for larger k and N.
    - Use of chi-square for small samples may decrease test power (higher chance of retaining false null hypothesis).

    - **Example (Large Sample)**: Training Need of Court Coordinators. H = 4.42. df = (3-1) = 2. Critical chi-square at α=0.05, df=2 is 5.991. Since 4.42 < 5.991, **null hypothesis is accepted**. No difference in training needs.
- **3.7 Comparison of ANOVA and Kruskal Wallis ANOVA Test**

    - **KW ANOVA**: Non-parametric equivalent of one-way ANOVA.
    - **ANOVA**: Compares **means** of different populations.
    - **KW ANOVA**: Compares **medians** of different populations.
    - **KW ANOVA**: Converts data into **ranks** before computation (known as "ANOVA by ranks").
    - **F-ratio as a function of H**: **F = [{(k-1)/(N-k)} {((N-1)/H)-1}]^-1**.

Unit 4: Chi-Square and Kendall Rank Correlation

- **4.2 Concept of Correlation**

    - **Definition**: Establishes relationships between two variables (X and Y).
    - **Direction**:
        - **Positive Correlation**: High X with high Y (or low X with low Y). Variables "go together".
        - **Negative Correlation**: High X with low Y (or vice versa). Variables move in opposite directions.
        - **Zero Correlation**: No typical trend, X values not predictive of Y.
    - **Range**: Correlation coefficients (r) range from **-1.00 to +1.00**.
        - +1.00: Perfect positive correlation.
        - -1.00: Perfect negative correlation.
        - 0.00: No relationship.
    - **Strength**: The closer r is to ±1.00, the stronger the relationship. Closer to 0.00, weaker the relationship.
    - **Scatter Plot**: Visualizes correlation. Reveals positive, negative, or no correlation patterns. Correlational strength cannot be quantified visually.
- **4.3 Measures of Correlation**

    - **Parametric Statistics**: **Pearson product-moment correlation coefficient (r)**.
    - **Non-parametric Statistics**:
        - **Spearman’s rank order correlation coefficient ("Spearman Rho")**: Assumes variables are measured on at least an **ordinal (rank order) scale**. Can be seen as Pearson 'r' computed from ranks.
        - **Kendall’s Tau (τ)**: Explained below.
        - **Chi-Square (X^2)**: For categorical variables.
- **4.4 Kendall’s Rank Order Correlation (Kendall’s tau): (τ)**

- A **descriptive statistical measure** of correlation/association for **rank-order data**, developed by Kendall (1938).
- **Null Hypothesis (Ho)**: In the population, the correlation between ranks of X and Y is **0 (τ = 0)**.
- **Purpose**: Evaluates the **degree of agreement between two sets of ranks** with respect to the relative ordering of all possible pairs of subjects/objects.
- **Concordant pairs**: Ranks in agreement (same sign/direction of difference).
- **Discordant pairs**: Ranks in disagreement (opposite sign/direction of difference).
- **τ = +1**: Complete agreement (all concordant).
- **τ = -1**: Complete disagreement (all discordant).
- **Comparison with Spearman's Rho (ρ)**:
  - Different underlying scales, **not directly comparable numerically**.
  - Both use same amount of information, equally likely to detect significant effect.
  - Rho is more commonly discussed (computations are more tedious for Tau).
  - Rho is like Pearson 'r' on ranks; Tau is the difference between probabilities of concordant and discordant pairs.
  - **Efficiency**: Both Rho and Tau have efficiency of 91% compared to Pearson 'r' (when 'r' is properly applicable).

- **4.5 Step by Step Procedure for Kendall Rank Order Correlation (without ties)**

  - Rank observations on X (1 to N) and Y (1 to N).
  - Arrange subjects so X ranks are in natural order (1, 2, 3, ... N).
  - Observe Y ranks in this new order.
  - Determine **S**: Number of agreements in order minus number of disagreements in order for Y ranks. (For each Y rank, count how many ranks to its right are larger (agreements) and how many are smaller (disagreements). Sum these counts for S.).
  - **Calculate T = 2S / [N(N-1)]**.
  - **For ties**: **T = 2S / [√(N(N-1)-Tx) * √(N(N-1)-Ty)]**.
    - Tx = Σt(t-1) for tied ranks on X; Ty = Σt(t-1) for tied ranks on Y.
  - **Test Hypothesis (X and Y independent)**:
    - N ≤ 10: Use Kendall's T table (Table I in some texts).
    - N > 10, < 30: Use critical value table for T.
    - N ≥ 30 (or for intermediate significance levels): **Calculate z = 3T * √[N(N-1) / (2(2N+5))]** and use the z-table. Reject Ho if probability is ≤ critical value.
- **4.7 Chi-Square Test (X^2)**

  - Measures the **alignment between two sets of frequency measures (categorical counts)**, not percentages or ratios.
  - Frequency numbers should be > 5 (generally).
  - **Functions**:

        - **Goodness of Fit**: Assesses if observed measures follow an expected pattern (e.g., from prior knowledge or average).
            - **Null Hypothesis (Ho)**: Observed and expected frequencies are **not significantly different**.
            - Attention on 0.05, 0.01, 0.001 figures.
        - **Measure of Independence**: Determines if two sets of measures **do not align** (i.e., are dependent).
            - **Null Hypothesis (Ho)**: Two sets of measures are **similar (independent)**.
            - Attention on 0.95 or 0.99 figures.
- **4.8 Relevant Background Information on Test**
    - **Chi-square goodness-of-fit test**: For single sample, categorizing n observations into k mutually exclusive categories.
    - Tests if there's a difference between **observed frequencies (O)** and **expected frequencies (E)**.
    - Significant result means observed frequency for at least one cell is not equal to expected.
    - **Assumptions**:
        - **Categorical nominal data** (frequencies for k mutually exclusive categories).
        - **Random sample of independent observations**.
        - **Expected frequency of each cell is 5 or greater** (some sources say no less than 1, and no more than 20% less than 5). If violated, cells can be combined.
    - **Robust test**: Still reliable even if assumptions are violated (e.g., for uniform distributions).
- **4.9 Step by Step Procedure for Chi-Square Test**
    - Write **Observed frequencies (O)**.
    - Figure **Expected frequencies (E)**:
        - Equal distribution: Total N / number of categories.
        - Prior knowledge: Total N * hypothesized population proportion.
    - **Calculate Chi-Square value (X^2)**: **X^2 = Σ [(O - E)^2 / E]**.
    - **Find df (degrees of freedom)**:
        - Goodness of fit: **df = N-1** (number of categories - 1).
        - Independence (contingency table): **df = (Number of Rows - 1) x (Number of Columns - 1)**.
    - **Find table value** (consult Chi-Square Table).
    - **Decision**:
        - If calculated X^2 **≥ table value**, **reject null hypothesis**. Significant difference.
        - If calculated X^2 **< table value**, **fail to reject null hypothesis**. Differences are likely due to chance.

    - **Example**: Car color preference. Calculated X^2 = 26.95. df = 4. Table X^2 at .05 for 4 df is 9.49. Since 26.95 > 9.49, **reject null hypothesis**. Real change in customer preferences.
- **4.10 Further Considerations about Chi-Square**

    - **Observations must appear in one cell only** (mutually exclusive).
    - **Actual frequencies** must be used, not percentages or means.
    - **Low Expected Frequencies**: A common limitation is that **expected frequencies should not fall below 5**. Cochran's rule (1954): No more than 20% of expected cells should be < 5. If violated, especially with small total samples (<20) and multiple cells < 5, risk of Type I error is too high.